

TYLER NORCROSS

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EDUCATION

University of New Hampshire
B.A. Computer Science, Algorithms

Durham, NH
May 2026

PROFESSIONAL EXPERIENCE

Agent Integrator

Founding Engineer

Remote
December 2025 – May 2026

- Served as **founding engineer** at an AI-native venture studio that embeds inside businesses to design and ship agentic systems & orchestration layers; oversaw architecture for **10+ client-facing** software products spanning health-tech, finance, and operations.
- Built full-stack internal operations platforms and public-facing web applications across multiple simultaneous **client engagements**; include API integrations, headless CMS portals, content generation workflows, chatbot tools, and more.
- Worked across a **modern tech stack** including Next.js, Node.js, TypeScript, PostgreSQL, Claude APIs, and third-party data APIs; contributed to multiple system architectures, visual design, production-grade code, prompt engineering for agentic workflows, and deployments.

TaskMind

Technical Co-Founder

Durham, NH
May 2025 – January 2026

- Co-founded** an AI-powered market intelligence SaaS platform; led full-stack architecture using Next.js, Node.js/Express, TypeScript, Google Cloud Platform (GCP) via App Engine, MongoDB; designed 50+ RESTful API endpoints with multi-tenant data isolation and JWT-based role access control; implemented end-to-end Stripe subscription infrastructure with tiered pricing and usage-based metering.
- Engineered a multi-model **AI orchestration layer** and a RAG enrichment pipeline integrating a dual-sanitization research system using Perplexity AI; features included vertical reporting, deep company research, competitive insights, and marketing campaigns.
- Awarded a place in the **Berkshire Innovation Center Stage 2 Accelerator (S2A)**; a competitive, equity-free 35-week program awarding a small cohort of early-stage tech startups a dedicated workspace, lab access, and a curated network of industry experts in Pittsfield, MA.

Dialogix

Full Stack Developer

Research Conference
April 2026

- Built a full-stack **AI conversation coaching platform** in Node.js/Express and React Native (Expo) with JWT + bcrypt authentication, per-user request throttling, and a MongoDB schema tracking conversation state with idempotent transitions to prevent data corruption on retries; implemented a psychometric personality profiling engine that infers defined core competencies.
- Engineered a **RAG pipeline** fusing Pinecone vector search (OpenAI text-embedding) with proprietary data from a sponsor-sanctioned Learning Management System (LMS); included a three-framework leadership model to generate context-aware coaching suggestions.
- Awarded **1st place** in the UNH Undergraduate Research Conference (computer science department); an annual celebration of academic excellence at UNH, and one of the largest conferences of its kind in the country.

PROJECTS

RoomNet

Roommate Matching Platform for University Housing

University Startup Competition
April 2024 & April 2025

- Built a **full-stack AI conversation coaching platform** in Node.js/Express and React Native (Expo) with JWT + bcrypt authentication, per-user request throttling, and a MongoDB schema tracking conversation state with idempotent transitions to prevent data corruption on retries; implemented a psychometric personality profiling engine that infers defined core competencies.
- Engineered a **RAG pipeline** fusing Pinecone vector search (OpenAI text-embedding) with proprietary data from a sponsor-sanctioned Learning Management System (LMS); included a three-framework leadership model to generate context-aware coaching suggestions.
- Placed twice in **Semifinals** for the UNH Holloway Competition; a premier business plan competition for undergraduate and graduate students. With over **\$40,000 in prizes**, it challenges students to develop products and present their plans to bring them to market.

Visual Design Agent

UI/UX Development Tool

Design Tool
April 2026 – Present

- Architected a **multimodal AI feedback loop** in Python that generates React/Tailwind components via Claude Haiku and hero images with OpenAI GPT Image 2 and renders them at 1440×900 and 390×844 via headless **Chromium (Playwright)**, and pipes screenshot bytes to Claude Sonnet for structured JSON visual critique; averaging 4 iterations to reach a reference quality threshold.
- Designed a 5-axis scoring system with precision fix instructions constrained to not regress high-scoring axes; deployed on **Hetzner VPS** (Ubuntu) with a 4-stage test suite validating each pipeline layer; every basis of visual reference is derived from professional templates.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, Java, C, SQL, React, Next.js, Node.js/Express, React Native/Expo, HTML/CSS

AI & Frontend: OpenAI API, Claude API, Perplexity API, RAG Pipelines, Prompt Engineering, LangChain, Pinecone, Tailwind CSS, Vite

Databases & Cloud: MongoDB, PostgreSQL, MySQL, SQLite, Firebase, Redis, GCP, Docker, CI/CD, GitHub Actions

Tools & APIs: REST, Stripe, Git/GitHub, Playwright, Prisma, Postman, Agile/Scrum, Linux/Unix, VPS, SSH